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Wikipedia as a Real-Time Proxy for CEO-Driven Shocks: Quantifying the Impact of Elon Musk's Digital Footprint on TSLA Returns

In the market-place, one convinces with gestures.
But reasons make the populace distrustful.

— Friedrich Nietzsche, Thus Spoke Zarathustra

Abstract

Financial markets often react sharply to the public behavior of high-profile executives, yet such attention is difficult to measure in a timely and structured way. This paper studies whether Wikipedia editing activity can serve as a real-time proxy for CEO-related attention in the case of Tesla and Elon Musk. We examine the editing histories of the Elon Musk and Tesla, Inc. pages and relate them to Tesla's daily stock returns while controlling for broader market movements. Our results show that Wikipedia editing activity of Elon Musk page is statistically significant in explaining Tesla's returns. In particular, one additional Wikipedia edit is associated with an average decrease of approximately 0.04% in Tesla's daily returns.

1 Introduction

Tesla provides an especially revealing setting for studying attention-driven market dynamics because the company's public image is unusually intertwined with that of its chief executive, Elon Reeve Musk. Over the past several years, Musk's statements, interviews, and online presence have repeatedly coincided with sharp movements in Tesla's stock price, suggesting that CEO visibility can itself become economically relevant. One widely cited example occurred on May 1, 2020, when Musk tweeted that "Tesla stock price is too high imo," after which Tesla shares fell by roughly 10% over the trading day. This makes Tesla a useful case for examining whether public attention directed at an individual executive can be separated from attention directed at the firm. At the same time, such attention is difficult to quantify in a timely and systematic way, since the underlying signals are dispersed across heterogeneous sources such as social media, news coverage, interviews, and public appearances.

In this setting, Wikipedia may provide a useful intermediate signal of public attention. Editing activity is generated by a relatively engaged group of users who actively monitor, revise, and update information in response to recent events. Although Wikipedia is not a direct measure of investor sentiment, it can reflect moments when public interest becomes sufficiently intense to trigger collective information production. This idea is consistent with prior research showing that Wikipedia activity can help predict real-world outcomes. For example, Mesty'an, Yasseri, and Kert'esz show that activity surrounding movie-related Wikipedia pages can predict box-office success before release [1]. More critically for financial applications, Moat et al. demonstrated that changes in how frequently Wikipedia users access and edit financially related pages can serve as an early indicator of stock market movements [2], establishing Wikipedia as a valid lens for investor attention.

Building on this intuition, this paper examines whether Wikipedia editing activity can serve as a real-time proxy for investor attention in Tesla's stock performance. We focus on two articles that are especially relevant to the company's public image and market dynamics: **Elon Musk** [3] and **Tesla, Inc.** [4].

The setting motivates two central research questions:

- Does greater editing intensity on the Elon Musk page correspond to lower Tesla stock returns?
- Which has the larger association with Tesla's returns: editing activity on the Elon Musk page or editing activity on the Tesla, Inc. page?

2 Data collection & description

The empirical analysis combines revision-level Wikipedia data with daily financial market data. On the Wikipedia side, we collected the complete edit history for two articles, *Elon Musk* [3] and *Tesla, Inc.* [4]. To do so, we developed a Python-based data collection pipeline that fetches the full revision metadata for a given article and stores it in a structured dataset. For each revision, the raw Wikipedia dataset contains the variables `revid`, `timestamp`, `user`, `size`, `sha1`, `flags`, `intro_text`, and `minor`. These fields make it possible to reconstruct the full revision history of each page and to derive time-varying measures of editing activity.

On the financial side, we collected daily time-series data for Tesla stock (TSLA) and for the S&P 500 index, using Python's library *yfinance*. We retained the closing price and trading volume as the key inputs for subsequent analysis. The main dependent variable is Tesla's daily log return,

$$returns_t := \log\left(\frac{close_t}{close_{t-1}}\right),$$

while the market control variable is the S&P 500 daily log return,

$$mkt_returns_t := \log\left(\frac{spclose_t}{spclose_{t-1}}\right).$$

These variables allow us to distinguish article-specific attention effects from broader market movements.

Because Wikipedia revisions arrive at irregular timestamps whereas financial variables are observed at the daily frequency, the raw revision-level metadata were aggregated to the day level before merging. From the full edit history of each article, we engineered three core daily Wikipedia features. First, $total_edits_t$ measures the total number of edits recorded on day t and serves as our primary proxy for editing intensity. Second, $unique_editors_t$ counts the number of distinct editors active on day t, capturing the breadth of participation around the article. Third, $reverts_t$ measures the number of revision events satisfying the condition $\texttt{sha1}_t = \texttt{sha1}_{t-2}$, which identifies simple reversal patterns and provides a proxy for conflict or correction dynamics in the editing process.

After constructing these daily Wikipedia measures separately for the two articles, we aligned them with the daily financial dataset by calendar date. The resulting analysis dataset therefore combines, for each day, Tesla stock outcomes, market controls, and article-specific attention measures derived from Wikipedia revision histories.

Raw Wikipedia revision-level variables:

Table 1: Revision-level Wikipedia metadata

Variable	Description	Type
`revid`	Unique identifier of the revision	Integer
`timestamp`	Date and time at which the revision was recorded	Datetime
`user`	Username or IP-based identifier of the editor	Text
`size`	Page size after the revision	Integer
`sha1`	Content hash used to track repeated article states	Text
`flags`	Revision flags returned by the Wikipedia API	Text
`intro_text`	Introductory article text associated with the revision snapshot	Text
`minor`	Indicator for whether the revision was marked as minor	Binary

Engineered daily variables used in analysis:

Table 2: Daily variables used in the merged analysis dataset

Variable	Description	Frequency
$returns_t$	Tesla daily log return computed from consecutive closing prices	Daily
$mkt_returns_t$	S&P 500 daily log return computed from consecutive index closes	Daily
mkt_volume_t	S&P 500 trading volume	Daily
$tsla_volume_t$	Tesla trading volume	Daily
$emsk_total_edits_t$	Total edits on the *Elon Musk* article on day t	Daily
$emsk_unique_editors_t$	Number of distinct editors active on the *Elon Musk* article on day t	Daily
$emsk_reverts_t$	Count of revert-type events on the *Elon Musk* article, defined by $\texttt{sha1}_t = \texttt{sha1}_{t-2}$	Daily
$tsla_total_edits_t$	Total edits on the *Tesla, Inc.* article on day t	Daily
$tsla_unique_editors_t$	Number of distinct editors active on the *Tesla, Inc.* article on day t	Daily
$tsla_reverts_t$	Count of revert-type events on the *Tesla, Inc.* article, defined by $\texttt{sha1}_t = \texttt{sha1}_{t-2}$	Daily

3 Data Analysis

The initial plot of daily edit frequency (Figure 1) of both articles over log of TSLA daily returns only motivates the first research question discussed before, as highly-edited periods often coincide with volatile stock market time windows.

3.1 Stationarity of TSLA log-returns

Moreover, the plot of log returns visually suggests *stationarity*, which, if formally confirmed, allows including $returns_{t-1}$ as explanatory variable in regression modeling steps. To assess this property more formally, we applied the Augmented Dickey–Fuller (ADF) test [5] to the Tesla return series. In its standard form, the test is based on the auxiliary regression. Also, the period after 2020 (inclusive) seems to be the most "interesting" in a sense of volatility of Wikipedia for Elon Musk and TSLA log-returns. That is why, for further research we will fix this time period.

$$\Delta returns_t = \alpha + \rho\, returns_{t-1} + \sum_{i=1}^{p} \phi_i \Delta returns_{t-i} + \varepsilon_t,$$

where $\Delta returns_t$ denotes the first difference of the return series and the lagged difference terms are included to control for serial correlation. The corresponding hypotheses are

$$H_0 : \rho = 0 \quad (unitroot/non-stationarity), \qquad H_1 : \rho < 0 \quad (stationarity).$$

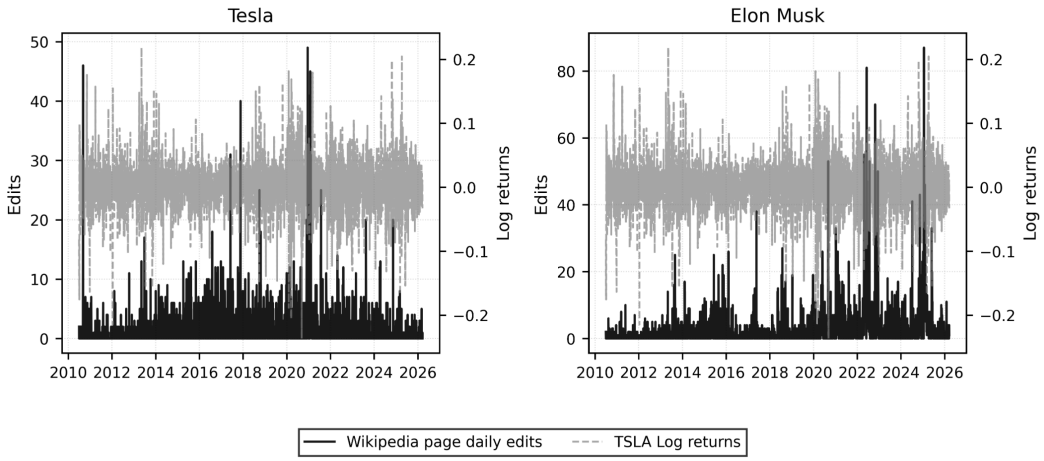

Figure 1: Daily edit frequency for the Elon Musk and Tesla, Inc. Wikipedia pages relative to Tesla daily log returns.

The test results may be accessed in Table 3. With the p-value of 0.0000 we can confidently reject the null hypothesis and confirm stationarity, which is really expected for log returns in stock markets. *Note: the stationarity is also confirmed for shortened time period after 2020.*

Table 3: Augmented Dickey–Fuller test results for Tesla return series

Statistic	**Value**
ADF test statistic	-63.2207
p-value	0.0000
Number of lags used	0
Number of observations	3955
1% critical value	-3.4320
5% critical value	-2.8623
10% critical value	-2.5672
Information criterion (IC best)	-14984.0330

3.2 Multicollinearity

Next, it is worth considering correlations between predictors to check for multicollinearity and choose a decent set of predictors. By looking at the correlation matrix in Figure 2, we can notice that *total_edits* seems to be highly correlated with *unique_editors* for both Tesla and Elon Musk articles. While *unique_editors* could serve as an interesting source of information, we decided to exclude it and let the coefficient near *total_edits* collect its effect because it is more intuitive to interpret. Noticeably, *emsk_reverts* seems to be more correlated with *emsk_total_edits*, meaning reverts take on a bigger portion of total edits for Musk's article than for Tesla's article. For now, it was decided not to exclude the possibility of including *reverts* and *total_edits* in a single regression equation. Pleasantly, no statistics extracted from Wikipedia pages seem to be strongly correlated across the two articles, allowing us later to include a mix of them in a single regression to address the second research question.

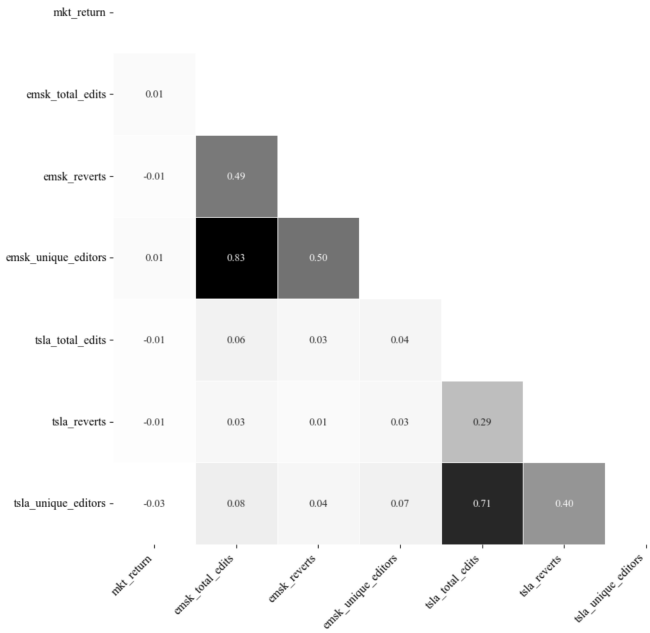

Figure 2: Correlation matrix of set of predictors.

4 Methods

4.1 Evaluation of CEO-Driven Shocks (Research Question 1)

To address the first research question, we have estimated an initial model to test the **Significance Hypothesis**. This stage investigates whether the aggregate editing intensity on the *Elon Musk* Wikipedia page has a non-zero association with Tesla's daily stock returns.

To maintain a balance between a sufficiently large sample size and the inclusion of the most market-relevant "digital shocks"—such as the acquisition of Twitter (X) and major CEO-led policy shifts—the model was estimated using data from **2020 to the present**. We utilize an Ordinary Least Squares (OLS) framework defined by the following specification:

$$log_return_t = \alpha + \beta_1 mkt_return_t + \beta_2 mkt_return_{t-1} + \beta_3 log_return_{t-1} + \beta_4 log_volume_{t-1} + \gamma_1 emsk_total_edits_t + \varepsilon_t$$

This specification is particularly suited for testing the first hypothesis for several reasons:

- **Isolation of Shocks**: By including both current and lagged S&P 500 returns, the model isolates article-specific attention from broader economic movements.
- **Momentum Control**: The inclusion of log_return_{t-1} ensures that the coefficient γ_1 does not capture pre-existing price trends or mean-reversion patterns.
- **Liquidity Control**: Integrating log_volume_{t-1} allows us to separate the Wikipedia-driven attention effect from general trading intensity.
- **Statistical Robustness**: To account for possible heteroscedasticity, we estimate the model using **HC1 robust standard errors**.

Under this framework, we tested the null hypothesis $H_0 : \gamma_1 = 0$ against the alternative $H_1 : \gamma_1 < 0$. Our preliminary results confirm that the *total_edits* variable on the Musk page is statistically significant, suggesting that intensified digital activity surrounding the CEO is associated with downward pressure on returns.

4.2 Magnitude of effect explained by Tesla article vs. by Elon Musk article (Research Question 2)

To address the second research question, we must include wikipedia metrics from both articles into the models. This setup is designed to determine whether the company's stock market is more sensitive to the digital footprint of it's executive or

the updates of the corporation. By including both Wikipedia proxies simultaneously, we isolate the marginal contribution of each signal.

The future estimating equation is defined as:

$$log_return_t = \alpha + \ldots + \gamma_1 emsk_edits_t + \gamma_2 tsla_edits_t + \varepsilon_t$$

The formal hypotheses for this comparative stage are:

- **Joint Significance**: $H_0 : \gamma_1 = \gamma_2 = 0$, testing whether both digital signals lack explanatory power when analyzed together.
- **Relative Dominance**: $H_0 : |\gamma_1| = |\gamma_2|$ vs. $H_1 : |\gamma_1| > |\gamma_2|$. This allows us to detect whether Elon Musk's public persona creates a dominant influence over the stock compared to the company's corporate identity.

5 Results

Table 4 presents the estimation results for our baseline OLS model (Research Question 1), utilizing HC1 robust standard errors to account for potential heteroscedasticity in the financial time series. The model demonstrates a solid explanatory power for daily financial data, yielding an R^2 of 0.203 and an overwhelmingly significant F-statistic (80.97, $p < 0.001$).

Table 4: OLS Regression Results: Impact of Elon Musk Wikipedia Edits on TSLA Returns

Variable	**Coefficient**	**Robust Std. Err.**	**z-statistic**	**P > |z|**
Intercept	0.0055	0.003	1.956	0.050
mkt_return	1.4878	0.082	18.064	0.000***
mkt_return_l1	0.1039	0.083	1.258	0.208
log_return_l1	0.0097	0.033	0.297	0.766
log_volume_l1	-0.0003	0.000	-1.979	0.048**
emsk_total_edits	**-0.0004**	**0.000**	**-2.607**	**0.009*****

Observations: 1,563 R^2: 0.203 Adj. R^2: 0.201
$^{**}p < 0.05$, $^{***}p < 0.01$

The primary variable of interest, *emsk_total_edits*, is statistically significant at the 1% level ($p = 0.009$). The estimated coefficient of -0.0004 indicates that, holding systematic market risk and historical momentum constant, one additional edit on the Elon Musk Wikipedia page is associated with a 0.04% decrease in Tesla's daily log returns. The control variables behave exactly as expected under standard asset pricing theory: the contemporaneous market return (*mkt_return*) acts as a robust proxy for systematic risk with a beta coefficient of approximately 1.49, while the lagged return (*log_return_l1*) is statistically insignificant, confirming the absence of strong serial momentum.

It is important to note that the magnitude of effect explained by Tesla article vs. by Elon Musk article (Research Question 2), which incorporates the *Tesla, Inc.* page metrics to test for relative dominance, has not yet been estimated in this interim phase of the project. The joint estimation of those signals remains a priority for the next stage.

6 Conclusions

This report provides empirical evidence that qualitative digital behavior, specifically, the editing intensity on Wikipedia can serve as a real-time proxy for CEO-driven market shocks. By isolating the revision history of the Elon Musk page from standard corporate equity controls, we successfully parameterized the concept of "narrative instability."

Our baseline regression confirms that heightened public attention directed at the chief executive carries distinct, negative economic weight. The strict statistical significance of the *total_edits* coefficient demonstrates that the market efficiently prices in the idiosyncratic risk generated by the CEO's public persona, translating digital controversy into quantifiable downward pressure on Tesla's stock. Wikipedia, acting as a filter requiring consensus and human effort, acts as a better alternative to raw social media noise in capturing true shifts in investor sentiment.

While our current specification successfully isolates this attention proxy, a significant portion of the variance naturally remains unexplained. Consequently, future extensions of this project will focus on deeply analyzing the residuals (ε_t) of our

current model. By examining the structure of this error term, we aim to identify and extract additional unobserved factors or omitted variables.

7 Works Cited

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